Decision Error

Grinnell College

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Strength of Evidence

So far, our process has been as follows:

1. Being with a null hypothesis, $H_0 : \mu = \mu_0$
2. Collected data and comptute statistic, i.e, $\overline{x}$
3. Compare our statistic against the null distribution, i.e., $t = \frac{\overline{x}-\mu_0}{\hat{\sigma}/\sqrt{n}}$
4. Derive a p-value based on the statistic and the distribution

We found that we could use our p-value to quantify the strength of evidence against our null: the smaller the p-value, the less likely our observed data if the null were true

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Decision Making

Based on the evidence we have collected, we must ultimately decide between one of two decisions:

1. There is sufficient evidence to reject H_0
2. There is *not* sufficient evidence to reject H_0

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Decision Making

Just as our confidence intervals were correct or incorrect, so to may be our decision regarding H_0. In this case, however, there are two distinct ways in which our decision can be incorrect:

1. H_0 is *TRUE*, yet we reject anyway
   - Example: our sample was outlier and resulted in large t-statistic
2. H_0 is *FALSE*, yet we fail to reject it
   - Example: our confidence interval was too large and contained the incorrect null hypothesis

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Decision Making

These two types of errors are known as Type I and Type II errors, respectively:

1. H_0 is *TRUE*, yet we reject anyway
   - Type I error
   - "False positive"
   - Evidence leads to wrong conclusion
2. H_0 is *FALSE*, yet we fail to reject it
   - Type II error
   - "False negative"
   - Not enough evidence to conclude

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Decision Making

	True State of Nature	
Test Result	H_0 True	H_0 False
Fail to reject H_0	Correct	Type II Error
Reject H_0	Type I Error	Correct

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Type I Errors

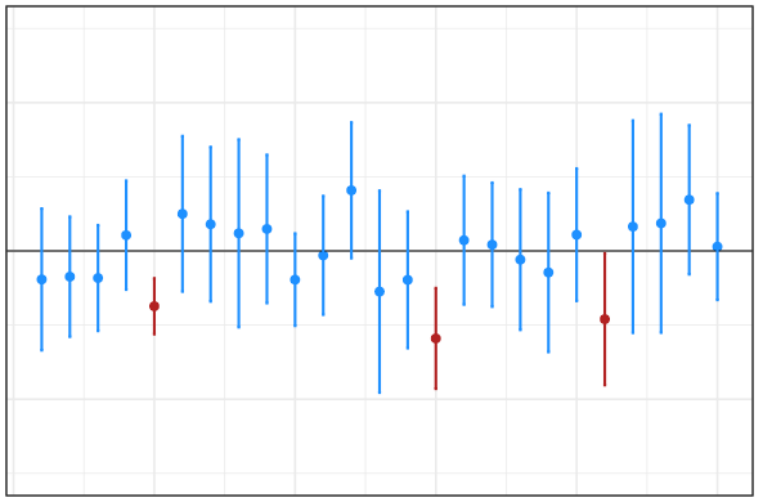
A Type I error describes a situation in which we incorrectly identify a null effect:

- Conclude that an intervention works when it does not
- Conclude that there is a relationship between two variables when there are not

A Type I error will occur, for example, when our null hypothesis is correct but our constructed confidence does not contain the true parameter (e.g., $\mu_0 = \mu$)

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Type I Errors



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Type I Error Rate

We can control the rate at which we commit Type I errors with adjusting the *level of significance*, denoted α.

This is also called the *Type I error rate*

The Type I error rate has a *one-to-one* correspondence with our confidence intervals: a 95% confidence interval will permit a Type I error 5% of the time, corresponding to $\alpha = 0.05$

We *reject* our null hypothesis when *p*-value $< \alpha$

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Type II Errors

A Type II error describes a situation in which the null hypothesis is false, yet based on the evidence gathered we fail to reject it:

- An intervention has a clinical effect, but it is not detected
- An email is considered spam, but the filter does not detect it

Typically, a Type II error is the result of one or more factors:

- Too few observations in our sample
- The population has large variability
- The effect size is small

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Effect Size

One important concept is identifying statistical differences is that of **effect size**, a value that measures the strength of association between two variables:

- Absolute difference between sample and hypothesis ($\overline{X} - \mu$)
- Standardized value of difference (t-statistic, z-score)
- Odds ratio, correlation, etc.,

Large effect sizes are much easier to detect, accommodating larger variances or smaller sample sizes. When the true effect size is small, more observations need to be collected to detect a difference

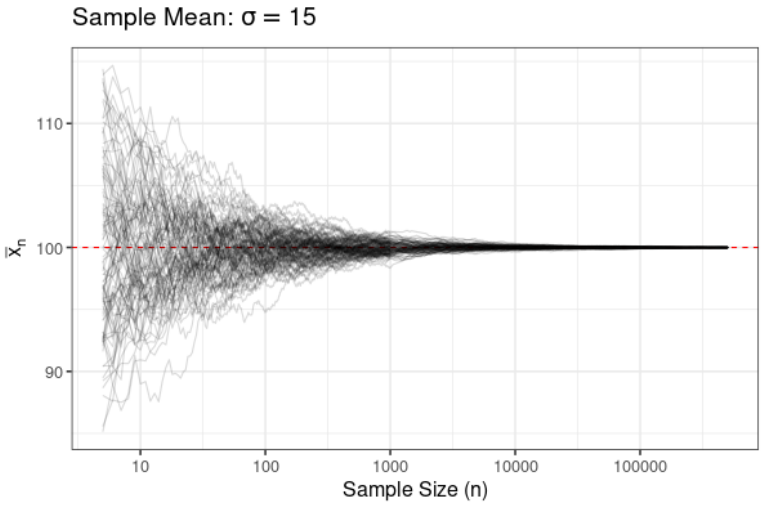
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Practical vs Statistical Significance

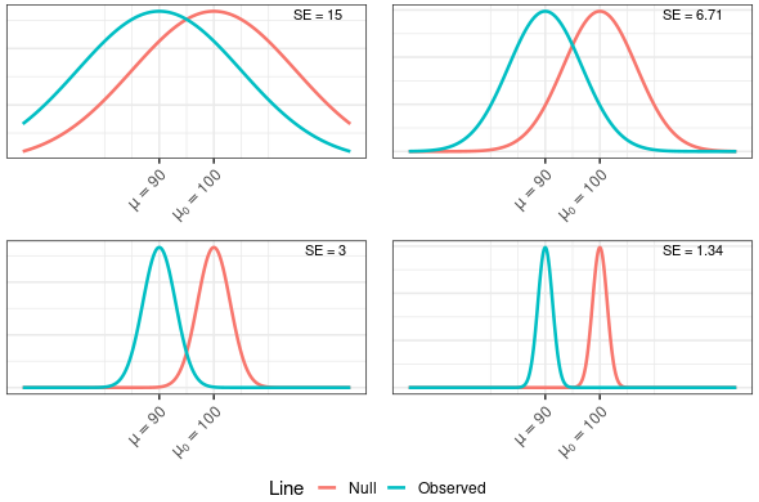
Suppose I have a coin and hypothesize that the probability of landing on heads is $H_0 : p = 0.5$. A collect a sample and find that $\hat{p} = 0.51$. Is this considered significant?

$\hat{p}$	n	t	p-value
0.51	100	0.400	0.6899
0.51	500	0.895	0.3713
0.51	1000	1.265	0.2060
0.51	10000	4.002	0.0001

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Type II Error Rate

The Type II error rate is typically denoted β

More frequently, we consider the rate at which Type II errors do not occur $(1 - \beta)$, a term we refer to as **power**

A study that is unable to detect a true effect is said to be **underpowered**

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Drawing Conclusions

As we never truly know whether H_0 is correct or not, we must simultaneously be prepared to combat both types of error

Test Result	True State of Nature	
	H_0 True	H_0 False
Fail to reject H_0	Correct $(1 - \alpha)$	Type II Error (β)
Reject H_0	Type I Error (α)	Correct $(1 - \beta)$

- Type I error $= P(\text{Reject } H_0 | H_0 \text{ true}) =$ false alarm
- Type II error $= P(\text{Fail to reject } H_0 | H_A \text{ true}) =$ missed opportunity

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